Software Design Specification
for
Kenakata- An E-Commerce Website

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Introduction

Purpose

A Software Design Specification (SDS) is a comprehensive document that describes how the software system will be designed and implemented to meet the "what" defined in the Software Requirements Specification (SRS). This Software Design Specification (SDS) document defines the technical blueprint for the development of the Kenakata E-Commerce Website.

It translates the requirements defined in the Software Requirements Specification (SRS) into a detailed design model that guides developers, testers, and maintainers throughout the system's lifecycle. The document defines the data structures, user interface designs, class diagram, use case diagram and extended use case diagram. The SDS ensures that the final product is consistent, maintainable, and scalable for future enhancement.

Intended Use

The SDS will be used for:

- Defining system components and their interactions.
- Describing data flow and information exchange between modules.
- Providing the database design and relationships.
- Establishing security mechanisms and system constraints.
- Supporting the testing phase through traceability between design and requirements.

Product Design Scope

The Kenakata E-Commerce System aims to deliver a robust and secure platform for online product sales and purchases. The design ensures seamless interactions between customers, sellers, and administrators through a unified web interface.

Major design goals include:

- Creating a user-friendly and responsive front-end for buyers and sellers.
- Implementing a secure payment gateway with proper encryption standards.
- Developing an admin dashboard to manage users, products, and orders efficiently.
- Ensuring scalability for growing product and user data.

Design Principles and Objectives

The design of Kenakata is guided by the following principles:

- Modularity: Each component operates independently and can be updated without affecting others.
- Reusability: Functions and components can be reused across different modules.
- Maintainability: Clear code structure and documentation for easy updates.
- Security: Robust encryption and user verification processes.
- Scalability: Architecture supports performance growth and user expansion.

Simple Design Plan for Kenakata Features

Customer (User) Features

User Registration & Authentication: We'll use a secure login process that requires an extra step, like a code to your phone, to keep accounts safe.

Product Search & Filtering: A specialized, fast search tool will be used to ensure all results appear almost instantly (in less than one second).

Product Details Viewing: Images and details will be loaded from a global network so they display very quickly, regardless of the user's location.

Add to Cart: Items placed in the cart will be temporarily saved in a super-fast memory system so the site always remembers them.

Add to Wishlist: Items saved here will be permanently linked to your profile so you can access them from any device.

Wishlist Sharing: You can share your list using a unique link that keeps your personal information private.

Checkout: This is a protected, step-by-step transaction that verifies inventory and final pricing before the payment stage.

Order Placement: The system automatically finalizes the order, instantly removes the item from stock, and marks the order status as "processing."

Payment Processing: We will use a flexible system that directs the payment to the correct service (card, mobile banking, or COD) based on your choice.

Mobile Banking Payment: The system will connect securely with the bank's service to process the payment and confirm it immediately.

Card Payment: We will use a secure third-party service to handle card payments, ensuring the system never stores your card number.

COD Payment: For Cash on Delivery, the system simply creates the order and flags it for the delivery team to collect money later.

Review & Rating: User reviews will be checked by an administrator before they are publicly displayed.

Return Process: A specific digital workflow will track your return request from submission until it receives final admin approval.

Refund Process: The system automatically communicates with the payment provider to send the money back to the user's original payment method.

Order Tracking: The system will regularly ask the delivery company for updates on your package's current location using the tracking number.

Live Chat Support: A separate, real-time messaging tool will be used to allow direct communication between customers and a support agent.

Notification System: The system will automatically send updates via email or text whenever something important happens, like an order status change.

Discount and Coupon Management: A special tool will validate coupon codes and calculate the final discounted price before the payment stage.

Delivery Partner Features

Delivery Partner Integration: This feature handles the necessary data exchange and system interface to manage and coordinate with external logistics services.

Order Tracking: Delivery partners utilize this to receive order details and update the status of the item during transit.

Admin Features:

Live Chat Support: Administrators need access to the live chat system to supervise, audit, or participate in support conversations.

Sales Report Generation: The administrator uses this function to create and view reports detailing sales performance, metrics, and trends.

Seller Monitoring: The administrator is responsible for overseeing the activities and performance of all registered sellers on the platform.

Notification System: The admin uses this to send mass or targeted announcements and alerts across the platform to users or sellers.

Seller Features:

Product Listing & Management: Sellers use this feature to add new products, modify existing listings, and update product information.

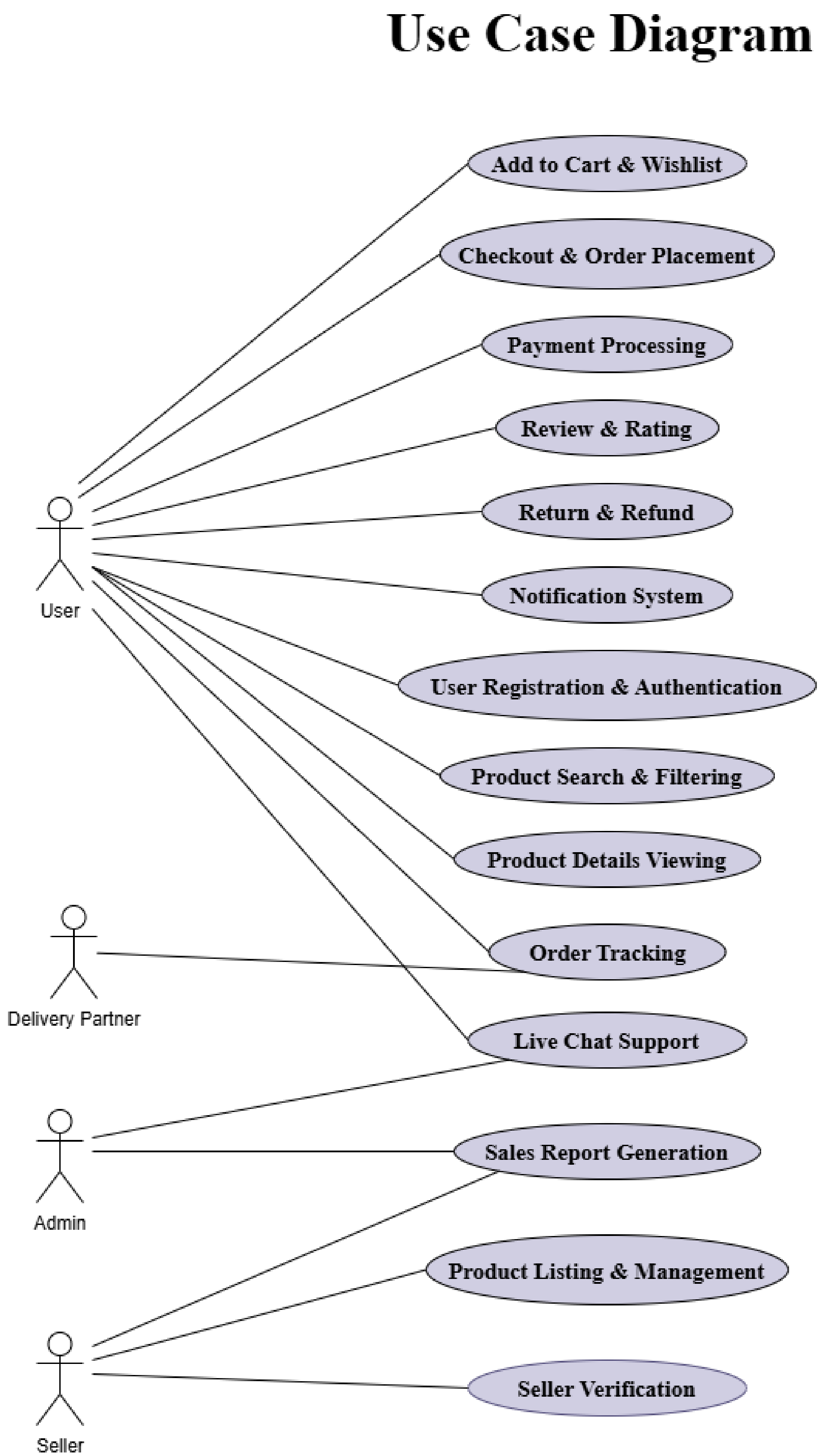
Inventory Management: This core function helps sellers keep track of stock levels for all listed products to prevent overselling.

Order Monitoring: Sellers must be able to view and manage all incoming and current orders pertaining to their products.

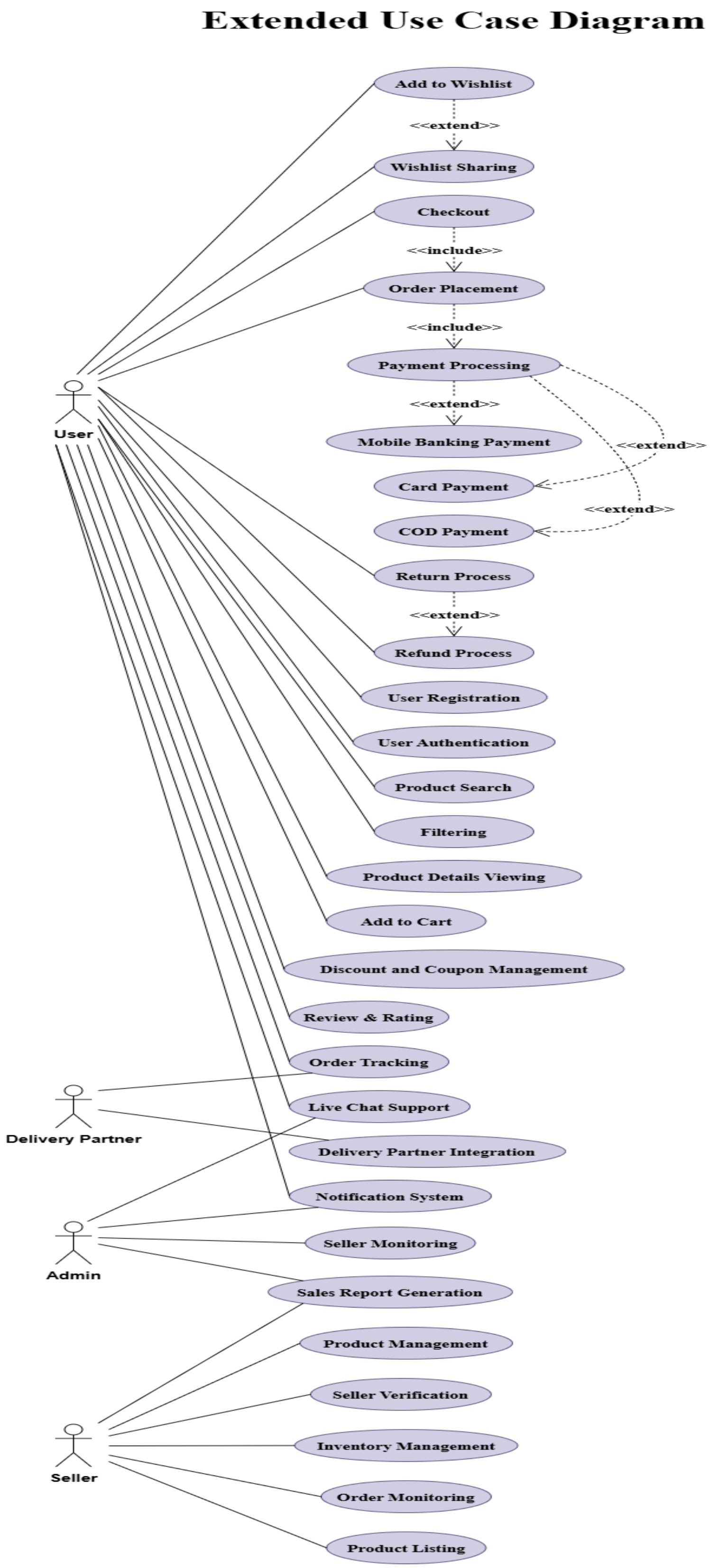
Seller Verification: The system needs a process for vetting and approving new sellers before they can list products.

Sales Report Generation: Sellers can access and generate reports detailing the performance and sales figures of their listed products.

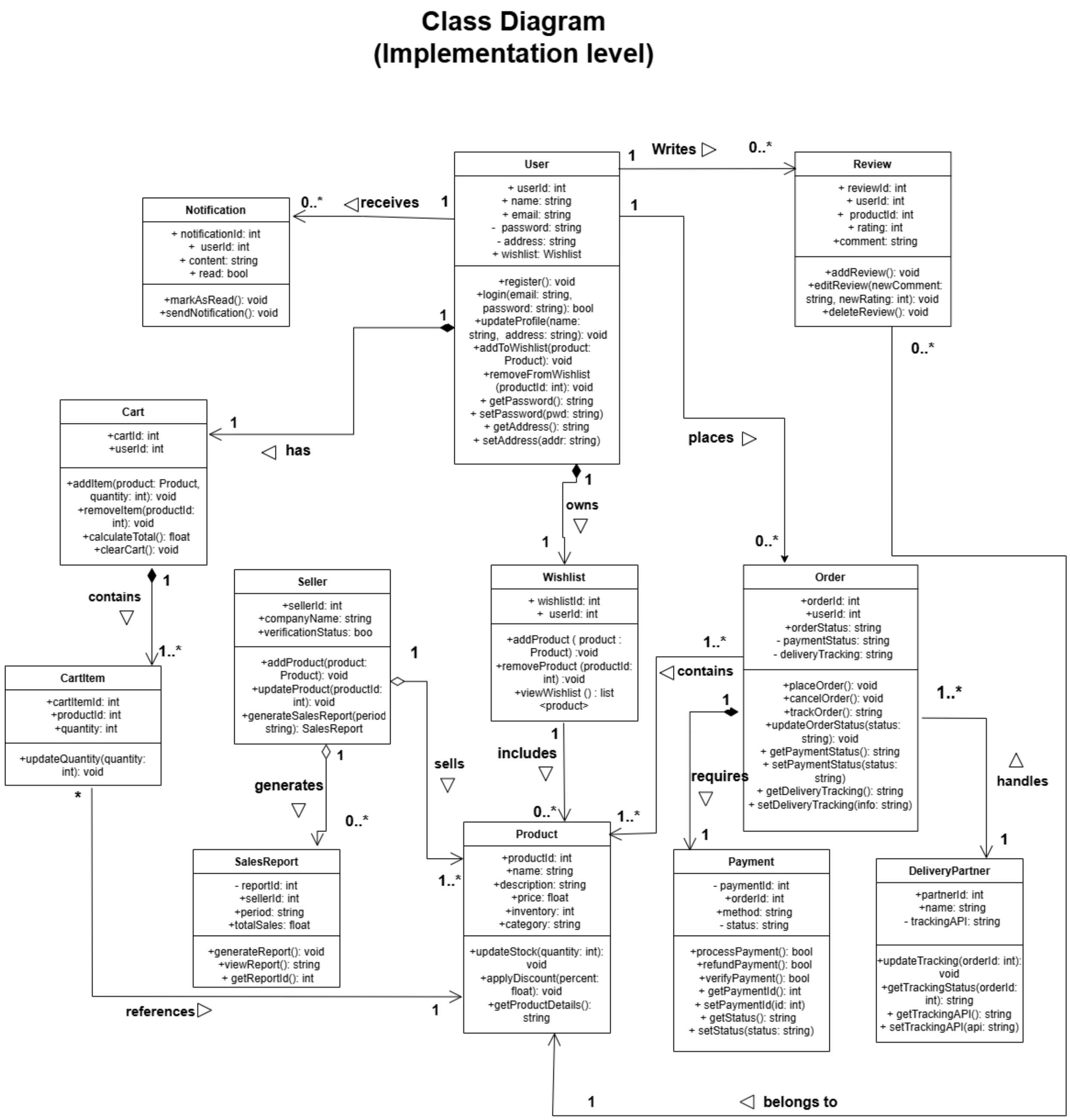
Use Case Diagram



Extended Use Case Diagram



Class Diagram (Implementation level)



The class diagram defines the object-oriented structure of the Kenakata E-Commerce System, showing how different classes interact to support system functionality.

1. User Class

- Represents customers or sellers using the system, containing personal details and associated wishlist.
- Methods handle registration, login, profile updates, password management, and wishlist operations.

2. Seller Class

- Extends user functionality for sellers, including company information and verification status.
- Manages product listings and generates sales reports.

3. Product Class

- Represents all product details like name, description, price, and category.
- Includes methods for updating stock, applying discounts, and retrieving detailed product information.

4. Cart Class

- Stores products temporarily before checkout, associated with a single user.
- Handles adding, removing, and calculating total prices for items in the cart.

5. CartItem Class

- Represents individual items in a cart with quantity and product references.
- Allows updating the quantity of each product in the cart.

6. Wishlist Class

- Contains products that users wish to buy later, linked to a specific user.
- Supports adding, removing, and viewing products in the wishlist.

7. Order Class

- Handles customer orders containing multiple products and linked to users and payments.
- Provides methods for placing, canceling, and tracking orders.

8. Payment Class

- Manages payment details for each order including amount, method, and status.
- Supports processing payments and tracking their state (successful or failed).

9. DeliveryPartner Class

- Represents third-party logistics partners responsible for delivery tracking.
- Updates and retrieves delivery information using tracking APIs.

10. Review Class

- Stores customer feedback for purchased products including ratings and comments.
- Provides methods to add, edit, or delete reviews associated with a user and product.

11. Notification Class

- Handles system messages and alerts sent to users about orders, offers, or updates.
- Supports sending and marking notifications as read.

12. SalesReport Class

- Maintains seller performance metrics such as total sales and report period.
- Generates and retrieves reports for business insights.

Class Relationships

User–Review (1 to 0..*): A user can write multiple reviews for different products.

User–Wishlist (1 to 1): Each user owns one wishlist.

User–Cart (1 to 1): Each user has one shopping cart.

Cart–CartItem (1 to many): A cart contains multiple cart items.

Order–Product (1 to many): Each order includes multiple products.

Order–Payment (1 to 1): Every order has exactly one payment transaction.

Order–DeliveryPartner (many to 1): Many orders can be handled by one delivery partner.

Seller–Product (1 to many): A seller can list multiple products.

Product–Review (1 to many): Each product can receive multiple user reviews.

Seller–SalesReport (1 to many): A seller can generate multiple reports.

Notification–User (many to 1): Multiple notifications can be sent to one user.

User Interface (UI) of each feature:

The User Interface (UI) Design for Kenakata – E-Commerce Website defines how users interact with the system visually and functionally. Each UI component ensures usability, accessibility, and consistent navigation across customer, seller, and admin panels.